

TP406: Geometry of Tensors and Applications

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Abstract. This is the lecture note of mine for a course on tensors and applications at DGIST (TP406) in 2026 Spring.

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We let $\mathbb{K}$ denote the base field of our spaces. In this lecture, mostly $\mathbb{K}$ will be either $\mathbb{C}$ or $\mathbb{R}$, and we will specify the base field if necessary. Moreover, in this lecture, all the vector spaces which we will deal with are finite dimensional. For a vector space V , we let V^* denote the dual space of V . We denote the vector space over k with basis $\{x_1, \dots, x_n\}$ by $\mathbb{K}\langle x_1, \dots, x_n \rangle$ and let $\mathbb{K}[x_1, \dots, x_n]$ be the polynomial ring over $\mathbb{K}$ with variables $x_1, \dots, x_n$.

For $d \in \mathbb{Z}_{>0}$, let $[d] := \{1, \dots, d\}$ and let $\mathfrak{S}_d$ denote the symmetric group on $[d]$ (or on d letters). Let $\mathbb{A}^n$ denote the affine space of dimension n , and $\mathbb{P}^n$ be the projective space of dimension n . For an affine (or a projective) variety X in an affine space (resp. a projective space), $\dim(X)$ denotes the dimension of X as an affine variety (resp. as a projective variety).

When we write an ideal (resp. affine or projective linear space) generated by a set S , we will write (S) (resp. $\langle S \rangle$). For a set S in a affine or projective space, we denote its Zariski closure by $\overline{S}$. For a set S in a projective space, $\widehat{S}$ means the affine cone of S . In addition, for a vector space V , we let $\mathbb{P}(V)$ denote the projectivization of V .

0. Introduction

Which things are we going to study during the course? Let's look at some key examples as follows.

0.1. Key examples

Example 1 (Complexity of Matrix Multiplication). Let $V := \text{Mat}_2(\mathbb{C})$ be the vector space of 2×2 complex matrices, and consider the matrix multiplication map $V \times V \rightarrow V$, $(A, B) \mapsto C = AB$, which will be denoted by $M_{(2)}$.

- (1) From the standard algorithm which we learned from linear algebra course, one can compute the output matrix

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix},$$

where we use 4 additions and 8 multiplications.

- (2) In 1969, V. Strassen found another way. In Strassen's algorithm, we can get the same output matrix C as

$$\begin{aligned} c_{11} &= M_1 + M_4 - M_5 + M_7, & c_{12} &= M_3 + M_5, \\ c_{21} &= M_2 + M_4, & c_{22} &= M_1 - M_2 + M_3 + M_6, \end{aligned}$$

where

$$M_1 = (a_{11} + a_{22})(b_{11} + b_{22}),$$

$$M_2 = (a_{21} + a_{22})b_{11},$$

$$M_3 = a_{11}(b_{12} - b_{22}),$$

$$M_4 = a_{22}(b_{21} - b_{11}),$$

$$M_5 = (a_{11} + a_{12})b_{22},$$

$$M_6 = (a_{21} - a_{11})(b_{11} + b_{12}),$$

$$M_7 = (a_{12} - a_{22})(b_{21} + b_{22}).$$

Note that here we need 18 additions and 7 multiplications.

Now, let's think about the multiplication of two $n \times n$ matrices. Say $n = 2^k$ (note that we can always get this situation just by adding some additional 0's to the original matrix). Using iterated multiplications of 2×2 block submatrices, in the standard algorithm we need to perform 8^k multiplications, whereas we have 7^k in the iterated Strassen's algorithm. It's also easy to see that the total number of arithmetic operations (that is, $+$ and $\cdot$) for this iterated $n \times n$ matrix multiplication is dominated by the number of multiplications. In this sense, we can say that to iterate Strassen's algorithm is much better than to use the standard one as n grows extensively.

- Exercise 1.* (1) Verify that Strassen algorithm produces the same output matrix C as the standard algorithm does.
- (2) Check that in the iterated $n \times n$ matrix multiplication (recall $n = 2^k$) the total number of arithmetic operations is dominated by the total number of multiplications (in other words, the total number of additions is a minor portion in the computation).

Based on the discussion above, we define an important complexity measure on matrix multiplication (M.M.) as follows. Let us denote the $n \times n$ matrix multiplication by $M_{\langle n \rangle}$.

Definition 0.1 (The exponent of matrix multiplication). The number

$$\omega := \inf_{\tau} \{ \tau \in \mathbb{R} : M_{\langle n \rangle} \text{ can be computed with } O(n^{\tau}) \text{ arithmetic operations} \}$$

is called the *exponent of matrix multiplication*.

In the standard algorithm, we have $8^k = n^{\tau} = (2^k)^{\tau}$, which means $\tau = 3$ and $\omega \geq 3$, while $\tau = \log_2 7$ and $\omega \geq 3 = \log_2 7 \approx 2.81$ in the Strassen's.

Computing or bounding the number ω is a flagship problem in theoretical computation complexity. As of January 2024, the world records for ω is ≤ 2.371339 ¹. We'll revisit this problem using 'tensor' language.

¹see also '[Computational complexity of matrix multiplication](#)' in Wikipedia

Example 2 (Power sum decomposition of polynomials). For a given polynomial, let us try to write it down as sum of power of a linear (i.e. degree 1) polynomials. For instance, we see

$$\begin{aligned} xy &= \frac{1}{2}(x+y)^2 - \frac{1}{2}(x-y)^2. \\ x^2y &= \frac{1}{3}(x+y)^3 - \frac{1}{3}(x-y)^3 - \frac{2}{3}(y)^3. \end{aligned}$$

What about $x^2y^2z^2$? It is also known (and you may find it still using elementary mathematics) that

$$\begin{aligned} x^2y^2z^2 &= \frac{1}{360} \{ (x+y+z)^6 + (x+y-z)^6 + (x-y+z)^6 + (x-y-z)^6 \\ &\quad - 2(x+y)^6 - 2(x-y)^6 - 2(x+z)^6 - 2(x-z)^6 - 2(y+z)^6 \\ &\quad - 2(y-z)^6 + 4x^6 + 4y^6 + 4z^6 \} . \end{aligned}$$

Now, for arbitrary polynomial $f(x_1, \dots, x_n)$, can we always find its power sum decomposition? If so, how to do this? This problem is also known as ‘Polynomial Waring problem’ following what E. Waring and other famous mathematicians did in number theory. Later, we’ll see how this problem can be related to a problem of tensors.

0.2. A brief review on basic linear algebra and More

Vector space, subspace, span, linear (in)dependence, linear maps, bases, representation matrices, dimension, product of vector spaces, and so on.. (will be updated later).

Quotient space Let V be a vector space over $\mathbb{K}$. For any $v \in V$ and a subspace $U \subset V$, first we set

$$v + U := \{v + u \in V : \forall u \in U\} .$$

We call $v + U$ an *affine subset* of V . Then, we define the quotient space V/U as

$$V/U := \{v + U : \forall v \in V\} .$$

From construction, we can see

Proposition 0.2. *The followings are all equivalent (TFAE):*

1. $v - w \in U$
2. $v + U = w + U$
3. $(v + U) \cap (w + U) \neq \emptyset$.

Now, let’s define

$$\begin{aligned} (v + U) + (w + U) &:= (v + w) + U \\ \lambda \cdot (v + U) &:= (\lambda \cdot v) + U . \end{aligned}$$

With these two basic operations, V/U forms a $\mathbb{K}$ -vector space. We call it the *quotient space of V by U* .

Proposition 0.3. *Suppose that V is a finite dimensional $\mathbb{K}$ -vector space and U is any subspace of V . Then,*

$$\dim(V/U) = \dim V - \dim U .$$

Exercise 2. 1. Prove Proposition 0.2.

2. Check that the quotient space $(V/U, +, \cdot)$ is indeed a $\mathbb{K}$ -vector space.
3. Prove Proposition 0.3 (hint: apply ‘Fundamental theorem of linear maps’ to $\pi : V \rightarrow V/U$).

Dual space and Dual map For any two $\mathbb{K}$ -vector spaces V, W , recall that

$$\text{Hom}_{\mathbb{K}}(V, W) := \{\text{all linear maps from } V \text{ to } W\}$$

has a $\mathbb{K}$ -vector space structure with natural addition and scalar multiplication of linear maps

$$(f + g)(x) := f(x) + g(x)$$

$$(\lambda \cdot f)(x) := \lambda \cdot f(x)$$

for any $f, g \in \text{Hom}_{\mathbb{K}}(V, W)$ and any $\lambda \in \mathbb{K}$ (we also use $\text{Hom}(V, W)$ simply when the base field is obvious).

In particular, when $W = \mathbb{K}$, we call $\text{Hom}(V, \mathbb{K})$ the *dual space of V* and usually denote it by V^* . Every $\phi \in V^*$ is said to be a *linear functional* on V . Thus, any linear functional ϕ on V gets a vector v in V and gives us back a scalar $\phi(v)$ in $\mathbb{K}$.

Here are some examples.

- Example 3.* 1. Let $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$ as $\phi(x, y, z) = 4x - 5y + z$. Then, ϕ is a linear functional on $\mathbb{R}^3$.
2. More generally, if $\phi : \mathbb{K}^n \rightarrow \mathbb{K}$ with $\phi(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n$ for any $c_i \in \mathbb{K}$, then $\phi \in \text{Hom}(V, \mathbb{K}) = V^*$.
 3. Set $\phi : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}$ with $\phi(p(x)) = \int_0^1 p(x)dx$. It’s easy to see that ϕ is also a linear functional on $\mathcal{P}(\mathbb{R})$.

Briefly speaking, by the duality one can think V (or V^*) is a mirror to the other. In principle, we can say anything about one vector space as we understand its dual.

Next, suppose that we have a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V . Then, its *dual basis* is a set $\{\phi_1, \dots, \phi_n\} \subset V^*$, where each ϕ_i is defined as $\phi_i(v_j) = 1$ if $i = j$ and $\phi_i(v_j) = 0$ otherwise.

Exercise 3. Check that $\{\phi_1, \dots, \phi_n\}$ is indeed a basis for V^* .

Corollary 0.4. *Assume that $\dim_{\mathbb{K}} V < \infty$. Then $\dim_{\mathbb{K}} V = \dim_{\mathbb{K}} V^*$.*

Remark 0.5. There are some remarks on dual spaces.

1. Usually, ϕ_i is denoted by v_i^* so that $v_i^*(v_j) = \delta_{ij}$.
2. (Double dual) As a $\mathbb{K}$ -vector space, there is a natural isomorphism $V \simeq (V^*)^*$. For any $v \in V$, one can interpret it as $v(\phi) := \phi(v)$ as an element of $(V^*)^*$.

Now, let us consider the dual of a linear map. For any $T \in \text{Hom}(V, W)$ and $\phi \in W^*$, one may take a composition $\phi \circ T$, which is an element of V^* . In this way, we have a natural map from W^* to V^* , which is indeed linear. Thus, we call the map $\phi \mapsto \phi \circ T$ the *dual map of T* and denote it by T^* . That is, $T^*(\phi) = \phi \circ T$ and $T^* \in \text{Hom}(W^*, V^*)$.

It is easy to see that the followings hold

1. $(S + T)^* = S^* + T^*$ for any $S, T \in \text{Hom}(V, W)$
2. $(\lambda \cdot T)^* = \lambda \cdot T^*$ for any scalar λ and $T \in \text{Hom}(V, W)$
3. $(S \circ T)^* = T^* \circ S^*$ for any $T \in \text{Hom}(U, V)$ and any $S \in \text{Hom}(V, W)$.

Finally, let's see how to compute the representation matrix of the dual map T^* . For any $T \in \text{Hom}(V, W)$, once we introduce ordered bases $\mathcal{A} = \{v_1, \dots, v_n\}$ for V and $\mathcal{B} = \{w_1, \dots, w_m\}$ for W , we get the representation matrix of T , $[T]_{\mathcal{B}}^{\mathcal{A}}$. Now, consider the dual map T^* and dual bases $\mathcal{A}^*, \mathcal{B}^*$. One natural question is 'how $[T^*]_{\mathcal{A}^*}^{\mathcal{B}^*}$ can be related to the original representation $[T]_{\mathcal{B}}^{\mathcal{A}}$?' The answer is

Exercise 4. Show that $[T^*]_{\mathcal{A}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{A}})^t$ (here M^t means the tranpose of M).

1. Tensor and Tensor Rank

Let $V_1, \dots, V_d$ be finite dimensional vector spaces over $\mathbb{K}$.

1.1. Tensor product

'Tensor' is a mathematical concept which captures *multi-linearity* (or *d-linearity*). So, one can say that tensors live everywhere multi-linearity does exist.

Mathematically, a *tensor* T is simply an element of a tensor product space $V_1 \otimes \dots \otimes V_d$. Tensor product is one of the most common ways to construct a new vector space from given ones : the *tensor product* of $V_1, \dots, V_d$, denoted by $V_1 \otimes \dots \otimes V_d$, is defined as the set of multi-linear functions from $V_1 \times \dots \times V_d \rightarrow \mathbb{K}$. Here, a map $\varphi : V_1 \times \dots \times V_d \rightarrow \mathbb{K}$ is said to be *multi-linear* (or *d-linearity*) if φ is linear with respect to each input, i.e. φ satisfies

$$\begin{aligned} \varphi(v_1 + c_1 v'_1, v_2, \dots, v_d) &= \varphi(v_1, v_2, \dots, v_d) + c_1 \varphi(v'_1, v_2, \dots, v_d); \\ \varphi(v_1, v_2 + c_2 v'_2, \dots, v_d) &= \varphi(v_1, v_2, \dots, v_d) + c_2 \varphi(v_1, v'_2, \dots, v_d); \\ &\vdots \\ \varphi(v_1, v_2, \dots, v_d + c_d v'_d) &= \varphi(v_1, v_2, \dots, v_d) + c_d \varphi(v_1, v_2, \dots, v'_d), \end{aligned}$$

for all $v_i, v'_i \in V_i$ and $c_i \in \mathbb{K}$. We can construct $V_1 \otimes \dots \otimes V_d$ as a quotient space of $\langle V_1 \times \dots \times V_d \rangle$ by a specific subspace described in the following. For simplicity, let me explain how $V_1 \otimes \dots \otimes V_d$ is defined by $d = 2$ case: $V_1 \otimes V_2$ is defined as the quotient space

$$\langle V_1 \times V_2 \rangle / L$$

where $\langle V_1 \times V_2 \rangle$ means the *free vector space* generated by the elements of $V_1 \times V_2$, which can be regarded as the vector space of all functions on $V_1 \times V_2$

that have only a finite number of nonzero values, and L is the subspace generated by the elements of the forms

$$\begin{aligned} (v_1 + v'_1, v_2) - (v_1, v_2) - (v'_1, v_2); \\ (v_1, v_2 + v'_2) - (v_1, v_2) - (v_1, v'_2); \\ (c_1 v_1, v_2) - c_1(v_1, v_2); \\ (v_1, c_2 v_2) - c_2(v_1, v_2), \end{aligned}$$

for any $v_1, v'_1 \in V_1, v_2, v'_2 \in V_2$ and each $c_1, c_2 \in \mathbb{K}$.

Similarly, we can construct the d -way tensor product space $V_1 \otimes \cdots \otimes V_d$ as the quotient space of the free vector space $\langle V_1 \times \cdots \times V_d \rangle$ by the corresponding subspace L (one may also understand $V_1 \otimes \cdots \otimes V_d$ as consecutive tensoring $(\cdots((V_1 \otimes V_2) \otimes V_3) \cdots \otimes V_d)$).

Now let us check some basic properties of $V_1 \otimes \cdots \otimes V_d$ and some related notions:

- (1) (Basis) Let $\{v_{i,1}, \dots, v_{i,m_i}\}$ be a basis for each V_i ,

$$\{v_{1,j_1} \otimes v_{2,j_2} \otimes \cdots \otimes v_{d,j_d} \mid 1 \leq j_1 \leq m_1, 1 \leq j_2 \leq m_2, \dots, 1 \leq j_d \leq m_d\}$$

is a basis for $V_1 \otimes \cdots \otimes V_d$.

- (2) (Dimension) $\dim(V_1 \otimes \cdots \otimes V_d) = (\dim V_1) \cdots (\dim V_d)$.

- (3) (Simple tensors) A tensor $T \in V_1 \otimes \cdots \otimes V_d$ is called (*totally*) *decomposable* or *simple* if $T = v_1 \otimes v_2 \otimes \cdots \otimes v_d$ where $v_i \in V_i$ for each $1 \leq i \leq d$. For example,

$$\begin{aligned} T &:= a_1 \otimes b_1 \otimes c_1 + a_1 \otimes b_1 \otimes c_2 + a_1 \otimes b_2 \otimes c_1 + a_1 \otimes b_2 \otimes c_2 \\ &= a_1 \otimes (b_1 + b_2) \otimes (c_1 + c_2) \end{aligned}$$

is a simple tensor in $A \otimes B \otimes C$, where $a_1 \in A, b_1, b_2 \in B$ and $c_1, c_2 \in C$.

- (4) (Universal property of Tensor product) From the construction, we have a following natural property of tensor product spaces:

“For any given d -linear map $F : V_1 \times \cdots \times V_d \rightarrow U$, there is a *unique* linear map $\tilde{F} : V_1 \otimes \cdots \otimes V_d \rightarrow U$ such that $F = \tilde{F} \circ \varphi$, where V_i and U are $\mathbb{K}$ -vector spaces, $\varphi : V_1 \times \cdots \times V_d \rightarrow V_1 \otimes \cdots \otimes V_d$ is a d -linear map with $\varphi(\sum(v_1, \dots, v_d)) = (\sum(v_1 \otimes \cdots \otimes v_d))$ and $\tilde{F}(v_1 \otimes \cdots \otimes v_d) = F(v_1, \dots, v_d)$.”

- (5) (Order and Format) For a tensor in $V_1 \otimes \cdots \otimes V_d$, d is called the *order* of the tensor, and $(\dim V_1, \dots, \dim V_d)$ is called the *format* of the tensor.
- (6) (Multi-dimensional arrays) If we fix bases of $V_1, \dots, V_d$, then there is one-to-one correspondence between a tensor in $V_1 \otimes \cdots \otimes V_d$ and a multidimensional array $[a_{i_1, \dots, i_d}]_{1 \leq i_1 \leq m_1, \dots, 1 \leq i_d \leq m_d}$ of size $(\dim V_1) \times \cdots \times (\dim V_d)$. See Figure 1.

1.2. Lesson from linear algebra

Since an order 2 tensor (i.e., when $d = 2$) corresponds to a matrix, it is an object in linear algebra. Note that

$$V_1 \otimes V_2 \cong \text{Hom}(V_1^*, V_2) \cong \text{Hom}(V_2^*, V_1) \cong \text{Hom}(V_1^* \otimes V_2^*, \mathbb{K}) = \text{Bilin}(V_1^* \times V_2^*, \mathbb{K})$$

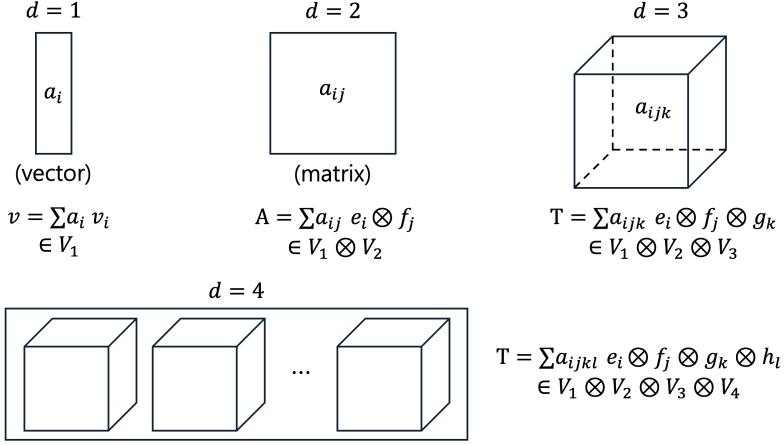


FIGURE 1. Tensors of order 1, 2, 3 and 4.

where $\text{Bilin}(V_1^* \times V_2^*, \mathbb{K})$ denote the set of bilinear functions from $V_1^* \times V_2^*$ to $\mathbb{K}$, and the last equality holds by *the universal property of tensor product* above.

Similarly, for order 3 and higher order tensors, we can exploit these natural identifications and interpretations. For instance, we have

$$\begin{aligned}
 U^* \otimes V^* \otimes W &= \text{Trilin}(U \times V \times W^*, \mathbb{K}) = (U \otimes V \otimes W^*)^* \\
 &= \text{Hom}(U \otimes V, W) = \text{Hom}(U, V^* \otimes W) \\
 &= \text{Hom}(U, \text{Hom}(V, W)) = \text{Hom}(V, U^* \otimes W) \\
 &= \text{Hom}(V, \text{Hom}(U, W)) = \dots
 \end{aligned}$$

and so on.

Example 4. (1) For $A = \mathbb{K}\langle a_1, a_2, a_3 \rangle$ and $B = \mathbb{K}\langle b_1, b_2 \rangle$, let

$$M = 3a_1^* \otimes b_1 + 2a_1^* \otimes b_2 - a_3^* \otimes b_1.$$

Note that M corresponds the linear map defined as

$$a_1 \mapsto 3b_1 + 2b_2, a_2 \mapsto 0, a_3 \mapsto -b_1$$

(2) Let $V = \mathbb{K}\langle e_1, e_2, e_3 \rangle$. The *identity map* $\text{Id}_V \in \text{Hom}(V, V) (= V^* \otimes V)$ can be represented as

$$\text{Id}_V = e_1^* \otimes e_1 + e_2^* \otimes e_2 + e_3^* \otimes e_3.$$

Moreover, the *trace map* $\text{tr} \in \text{Bilin}(V \times V^*, \mathbb{K}) (= V^* \otimes V)$ which is defined as $\text{tr}(v, \alpha) = \alpha(v)$ and by extending bilinearly can be represented as

$$\text{tr} = \sum_{i=1}^3 e_i^* \otimes e_j \otimes \delta_{ij} = e_1^* \otimes e_1 + e_2^* \otimes e_2 + e_3^* \otimes e_3.$$

Thus, Id_V is same as tr as tensor.

Let $M \in V_1 \otimes V_2$ be a matrix, and say $\dim V_1 = a$ and $\dim V_2 = b$. Note that, using $GL(V_1) \times GL(V_2)$ -action, we may transform M into

$$\left[\begin{array}{c|c} \begin{matrix} 1 & & \\ & \ddots & \\ & & 1 \end{matrix} & O \\ \hline O & O \end{array} \right]$$

where upper left block is $r \times r$ for $r = \text{rank } M$. (It is a *rank decomposition* of M .) In other words,

$$M = a_1 \otimes b_1 + \cdots + a_r \otimes b_r$$

for some $a_1, \dots, a_r \in V_1$ and $b_1, \dots, b_r \in V_2$.

In addition, from the $GL(V) \subset GL(V) \times GL(V)$ -action defined conjugation, i.e., for $P \in GL(V) \subset GL(V) \times GL(V)$, $P \cdot M = PMP^{-1}$, M can be represented by block/triangular/diagonal matrix.

We have the following fundamental results about the rank of M :

$$\begin{aligned} \text{rank}(M) &= \min\{r : \text{all } (r+1)\text{-minors of } M \text{ vanish}\} \\ &= \dim(\text{Im}(M : V_1^* \rightarrow V_2)) \\ &= \dim(\text{Im}(M : V_2^* \rightarrow V_1)) \\ &= \min\{r : M = UW \text{ where } U \text{ is an } a \times r \text{ matrix and } W \text{ is an } r \times b \text{ matrix}\} \end{aligned}$$

See Figure 2. Note that the M 's on the second and third lines above are

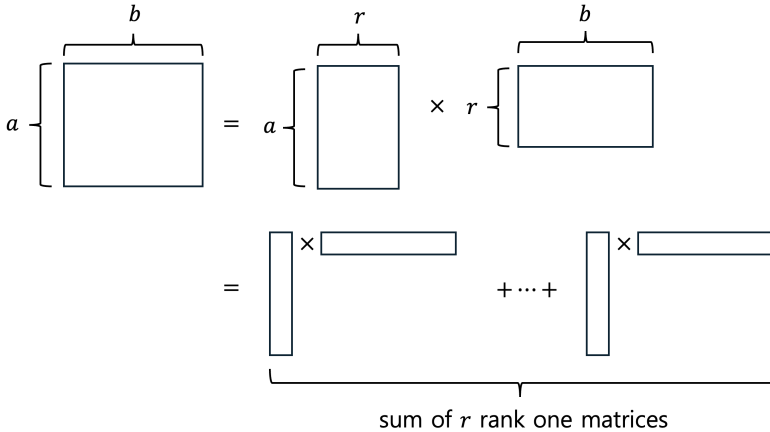


FIGURE 2. Rank decompositions of matrices.

“transpose” to each other.

1.3. Tensor decomposition and tensor rank

We would like to generalize these notions and results in the previous section into the case of d -way tensors $V_1 \otimes \cdots \otimes V_d$.

Definition 1.1 (Tensor rank). For $T \in V_1 \otimes \cdots \otimes V_d$,

$$\mathbf{R}(T) := \min\{r : T = \sum_{i=1}^r v_{1,i} \otimes v_{2,i} \otimes \cdots \otimes v_{d,i}; v_{j,i} \in V_j \text{ for each } 1 \leq j \leq d\}$$

is called the *tensor rank* of t . The expression $T = \sum_{i=1}^r v_{1,i} \otimes v_{2,i} \otimes \cdots \otimes v_{d,i}$ is called a *tensor decomposition* (or *CP decomposition*).

The ‘celebrated’ SVD (over $\mathbb{R}$ or $\mathbb{C}$) always gives a ‘good’ rank one decomposition

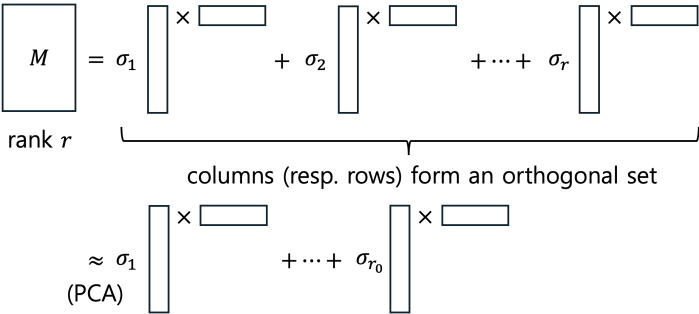
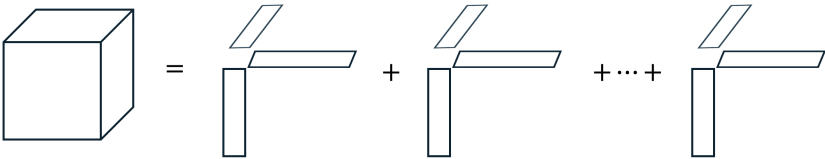


FIGURE 3. SVD and PCA.

A tensor decomposition



→ There are various tensor versions of PCA

FIGURE 4. Figure of tensor decomposition.

The followings are big problems about tensor:

- (1) Compute(or bound) the rank of a given tensor.
- (2) Find the number of decompositions of a specific/general tensor. (This problem is related with the number of “fibers” of a secant variety.)

- (3) Approximate given tensors with tensors of low rank (as we did for matrices using SVD).

Recall that above (1)-(3) on matrices(i.e., order 2 tensors) have been studied extremely. However, for order ≥ 3 tensors, (1)-(3) are widely open.

2. Examples and Basic Properties

2.1. Data analysis

Example 5 (Tensor from statistics). Let

$$A = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 6 & 12 & 18 & 24 \\ 14 & 28 & 42 & 56 \\ 30 & 60 & 95 & 120 \end{bmatrix}.$$

We can decompose A as

$$A = \begin{bmatrix} 1 \\ 3 \\ 7 \\ 15 \end{bmatrix} [2 \quad 4 \quad 6 \quad 8] \text{ (and hence } \text{rank}(A) = 1).$$

From this decomposition, we may obtain $A = [a_{ij}]$ satisfies

$$a_{ij} = 2j \cdot (2^i - 1).$$

Now, let

$$B = \begin{bmatrix} 15 & 13 & 9 & 1 \\ 47 & 45 & 37 & 17 \\ 81 & 83 & 75 & 47 \\ 119 & 133 & 133 & 105 \end{bmatrix}.$$

We can decompose B as

$$B = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 14 \end{bmatrix} [1 \quad 3 \quad 5 \quad 7] + \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix} [15 \quad 13 \quad 9 \quad 1] \text{ (and hence } \text{rank}(A) = 2).$$

From this decomposition, we may obtain $B = [b_{ij}]$ satisfies

$$b_{ij} = (2^i - 2)(2j - 1) + (2i - 1)(17 - 2j).$$

As we have seen with these examples, tensor decomposition is important for analyzing data.

2.2. 3-way tensors

Example 6 (Outer product in $\mathbb{R}^3$). Let $V = \mathbb{R}^3$. Consider the *outer product map*

$$\mu : V \times V \rightarrow V, (\mathbf{a}, \mathbf{b}) \mapsto \mathbf{a} \times \mathbf{b}.$$

Note that μ is a bilinear map, and so $\mu \in V^* \otimes V^* \otimes V (\cong \mathbb{R}^{27})$. Take a standard basis $\{e_1, e_2, e_3\}$ of V . Consider its dual basis $\{e_1^*, e_2^*, e_3^*\}$. Then we obtain

$$\begin{aligned} \mu &= e_2^* \otimes e_3^* \otimes e_1 - e_3^* \otimes e_2^* \otimes e_1 \\ &\quad - e_1^* \otimes e_3^* \otimes e_2 + e_3^* \otimes e_1^* \otimes e_2 \\ &\quad + e_1^* \otimes e_2^* \otimes e_3 - e_2^* \otimes e_1^* \otimes e_3. \end{aligned}$$

from the definition of outer product: for $\mathbf{a} = a_1e_1 + a_2e_2 + a_3e_3$ and $\mathbf{b} = b_1e_1 + b_2e_2 + b_3e_3$,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} e_1 & e_2 & e_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

Thus, $\mathbf{R}(\mu) \leq 6$.

Now, we have a natural question, $\mathbf{R}(\mu) = 6$ or $\mathbf{R}(\mu) \leq 5$?

Proposition 2.1. *Let A, B and C be vector spaces. For any $T \in A \times B \times C \cong \text{Hom}(A^*, B \otimes C)$,*

$$\mathbf{R}(T) = \min\{r : T(A^*) \subset \langle b_1 \otimes c_1, \dots, b_r \otimes c_r \rangle \text{ in } B \otimes C\}$$

where $T(A^*)$ denotes the image of $T : A^* \rightarrow B \otimes C$.

Corollary 2.2. *Say $\dim A = a, \dim B = b, \dim C = c$ and $a \leq b \leq c$. For any $T \in A \otimes B \otimes C$, we have $\mathbf{R}(T) \leq a \cdot b$.*

Note that this upper bound is not sharp in general.

Proof of Proposition 2.1. We prove by showing two inequalities.

- ($\geq$) Suppose that $\mathbf{R}(T) = r$, i.e., t can be represented by $T = \sum_{i=1}^r a_i \otimes b_i \otimes c_i$. Then $T(A^*) \subset \langle b_1 \otimes c_1, \dots, b_r \otimes c_r \rangle$. Hence, $\mathbf{R}(T) \geq \min\{r : T(A^*) \subset \langle b_1 \otimes c_1, \dots, b_r \otimes c_r \rangle \text{ in } B \otimes C\}$.
- ($\leq$) On the other hand, say $T(A^*)$ is spanned by $b_1 \otimes c_1, \dots, b_r \otimes c_r$. Let $A = \mathbb{K}\langle a_1, \dots, a_s \rangle$ and $A^* = \mathbb{K}\langle a_1^*, \dots, a_s^* \rangle$. Then $T(a_i^*) = \sum_{j=1}^r \lambda_j^i \cdot b_j \otimes c_j$ for all $i = 1, \dots, s$. Hence, $T = \sum_{i,j} a_i \otimes (\lambda_j^i \cdot b_j \otimes c_j) = \sum_{j=1}^r (\sum_{i=1}^s \lambda_j^i a_i) \otimes b_j \otimes c_j$. Thus, $\mathbf{R}(T) \leq \min\{r : T(A^*) \subset \langle b_1 \otimes c_1, \dots, b_r \otimes c_r \rangle \text{ in } B \otimes C\}$.

□

For an order 3 tensor $t \in A \otimes B \otimes C$, it is sufficient to check how many rank 1 matrices in $B \otimes C$ we need for describing $t(A^*)$, in order to find $\mathbf{R}(t)$.

Example 7 (Outer product in $\mathbb{R}^3$ again). Recall that the outer product tensor μ is an element in $V^* \otimes V^* \otimes V$ ($V = \mathbb{R}^3$). Consider $\mu(V^*) \subset V^* \otimes V^* =$

$\text{Hom}(V, V^*)$, where the V^* in μ corresponds to the last V on $V^* \otimes V^* \otimes V$. Note that we can take

$$\begin{aligned} e_2^* \otimes e_3^* &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad -e_3^* \otimes e_2^* = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \\ -e_1^* \otimes e_3^* &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad e_3^* \otimes e_1^* = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\ e_1^* \otimes e_2^* &= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad -e_3^* \otimes e_2^* = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Exercise 5. Show that the outer product tensor $\mu \in V^* \otimes V^* \otimes V$ ($V = \mathbb{R}^3$) has tensor rank at most 5. (Hint: find 5 rank one matrices $A_1, \dots, A_5$ such that $\mu(V^*) \subset \langle A_1, \dots, A_5 \rangle$ in $V^* \otimes V^*$.)

Example 8 (W-state tensor). The tensor

$$W = e_0 \otimes e_0 \otimes e_1 + e_0 \otimes e_1 \otimes e_0 + e_1 \otimes e_0 \otimes e_0 \in \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2,$$

where $\{e_1, e_2\}$ is a standard basis of $\mathbb{C}^2$, is called the *W-state tensor*. From its formula, we obtain that $\mathbf{R}(W) \leq 3$. Now, we can ask a natural question: $\mathbf{R}(W) = 3$ or $\mathbf{R}(W) \leq 2$? In fact, $\mathbf{R}(W) = 3$. We can prove it considering a system of polynomial equation. If W has a decomposition of the form

$$\begin{aligned} W &= (a_0 e_0 + a_1 e_1) \otimes (b_0 e_0 + b_1 e_1) \otimes (c_0 e_0 + c_1 e_1) \\ &\quad + (d_0 e_0 + d_1 e_1) \otimes (\lambda_0 e_0 + \lambda_1 e_1) \otimes (f_0 e_0 + f_1 e_1), \end{aligned}$$

then we get the system of 8 cubic equations in 12 unknowns

$$\begin{cases} a_0 b_0 c_0 + d_0 \lambda_0 f_0 = 0 \\ a_0 b_0 c_1 + d_0 \lambda_0 f_1 = 1 \\ a_0 b_1 c_0 + d_0 \lambda_1 f_0 = 1 \\ a_0 b_1 c_1 + d_0 \lambda_1 f_1 = 0 \\ a_1 b_0 c_0 + d_1 \lambda_0 f_0 = 1 \\ a_1 b_1 c_0 + d_1 \lambda_1 f_0 = 0 \\ a_1 b_0 c_1 + d_1 \lambda_0 f_1 = 0 \\ a_1 b_1 c_1 + d_1 \lambda_1 f_1 = 0 \end{cases}. \quad (2.1)$$

That is, the problem is reduced to proving inconsistency of the system (2.1).

Exercise 6. Verify the W-state tensor W has rank 3, i.e., show the inconsistency of the system (2.1).

There is a classical result about tensor rank of 3-way tensors:

Theorem 2.3 (B. Segre). *A “general” tensor over $\mathbb{C}$ of format $(2, 2, 2)$ has a unique decomposition as a sum of two decomposable tensors.*

Remark 2.4 (Fundamental theorem of linear algebra). Let V, W be vector spaces, and let $f : V \rightarrow W$ be a linear map. Then the followings hold:

(1) We have

$$\begin{aligned} \text{rank}(M) &= \min\{r : \text{all } (r+1)\text{-minors of } M \text{ vanish}\} \\ &= \dim(\text{Im}(M : V_1^* \rightarrow V_2)) \\ &= \dim(\text{Im}(M : V_2^* \rightarrow V_1)) \\ &= \dim(V) - \dim(\text{Ker}(f)). \end{aligned}$$

In particular, $\text{rank}(f) \leq \min\{\dim(V), \dim(W)\}$.

- (2) $\text{rank}(f) \leq r$ if and only if all the $(r+1)$ -minors of the matrix M_f corresponding (in any basis) to f are vanishing.
- (3) For “general” $f \in V^* \otimes W$, $\text{rank}(f) = \min\{\dim(V), \dim(W)\}$ (i.e., of full rank).
- (4) If a sequence $\{f_t\}$ of linear maps of rank r having a limit f_0 , then $\text{rank}(f_0) \leq r$. (It holds since the set of linear maps of rank $\leq r$ is Zariski closed by (2), furthermore, Euclidean closed in this case.)

However, these properties are no more true for tensors of order ≥ 3 . In particular, the set of tensors of tensor rank $\leq r$ is neither open nor closed in general.

3. Border Rank and Complexity

3.1. Border rank

Let $\{M_\epsilon\}$ be a family of matrices of rank r for any $\epsilon > 0$. Recall that

$$\text{rank}(\lim_{\epsilon \rightarrow 0} M_\epsilon) \leq r.$$

However, for a family of tensors $\{T_\epsilon\}$ with $\mathbf{R}(T_\epsilon) = r$ for any $\epsilon > 0$, all of

$$\mathbf{R}(\lim_{\epsilon \rightarrow 0} T_\epsilon) < r, \quad \mathbf{R}(\lim_{\epsilon \rightarrow 0} T_\epsilon) = r, \quad \text{and} \quad \mathbf{R}(\lim_{\epsilon \rightarrow 0} T_\epsilon) > r$$

can happen.

Example 9. Recall that the W-state tensor

$$W = e_0 \otimes e_0 \otimes e_1 + e_0 \otimes e_1 \otimes e_0 + e_1 \otimes e_0 \otimes e_0 \in \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2,$$

where $\{e_1, e_2\}$ is a standard basis of $\mathbb{C}^2$, satisfies $\mathbf{R}(W) = 3$. However,

$$W_\epsilon := \frac{1}{\epsilon} \{(e_0 + \epsilon e_1) \otimes (e_0 + \epsilon e_1) \otimes (e_0 + \epsilon e_1) - e_0 \otimes e_0 \otimes e_0\}$$

satisfies $\mathbf{R}(W_\epsilon) = 2$ for any $\epsilon > 0$ and $\lim_{\epsilon \rightarrow 0} W_\epsilon = W$.

Let $V_1, \dots, V_d$ be vector spaces.

Definition 3.1 (Border rank). For any $T \in V_1 \otimes \dots \otimes V_d$,

$$\underline{\mathbf{R}}(T) := \min \left\{ r : \begin{array}{l} \text{there exists a family } \{T_\epsilon\} \text{ such that} \\ \mathbf{R}(T_\epsilon) = r \text{ for all } \epsilon > 0 \text{ and } \lim_{\epsilon \rightarrow 0} T_\epsilon = T \end{array} \right\}$$

is called the *border rank* of T .

From the definitions of tensor rank and border rank, we obtain

$$\underline{\mathbf{R}}(T) \leq \mathbf{R}(T)$$

for every tensor T .

Exercise 7. Find examples of tensors $T_\epsilon \rightarrow T_0 \neq 0$ such that for any $\epsilon \neq 0$, $\mathbf{R}(T_\epsilon)$ is constant, but

- (1) $\mathbf{R}(T_0) < \mathbf{R}(T_\epsilon)$;
- (2) $\mathbf{R}(T_0) = \mathbf{R}(T_\epsilon)$;
- (3) $\mathbf{R}(T_0) > \mathbf{R}(T_\epsilon)$.

3.2. Complexity

Let $V := \text{Mat}_2(\mathbb{C})$ be the vector space of 2×2 complex matrices, and consider matrix multiplication map $V \times V \rightarrow V$, $(A, B) \mapsto C = AB$, which is a tensor denoted by $M_{\langle 2 \rangle}$ in $V^* \otimes V^* \otimes V$. Let us think of $\mathbf{R}(M_{\langle 2 \rangle})$.

- (1) From the standard algorithm

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix},$$

we obtain the formula

$$\begin{aligned} M_{\langle 2 \rangle} &= a_{11}^* \otimes b_{11}^* \otimes c_{11} + a_{12}^* \otimes b_{21}^* \otimes c_{11} \\ &\quad + a_{21}^* \otimes b_{11}^* \otimes c_{21} + a_{22}^* \otimes b_{21}^* \otimes c_{21} \\ &\quad + a_{11}^* \otimes b_{12}^* \otimes c_{12} + a_{12}^* \otimes b_{22}^* \otimes c_{12} \\ &\quad + a_{21}^* \otimes b_{12}^* \otimes c_{22} + a_{22}^* \otimes b_{22}^* \otimes c_{22}. \end{aligned}$$

Hence, $\mathbf{R}(M_{\langle 2 \rangle}) \leq 8$.

- (2) From Strassen's algorithm

$$\begin{aligned} c_{11} &= M_1 + M_4 - M_5 + M_7, \quad c_{12} = M_3 + M_5, \\ c_{21} &= M_2 + M_4, \quad c_{22} = M_1 - M_2 + M_3 + M_6, \end{aligned}$$

where

$$\begin{aligned} M_1 &= (a_{11} + a_{22})(b_{11} + b_{22}), \\ M_2 &= (a_{21} + a_{22})b_{11}, \\ M_3 &= a_{11}(b_{12} - b_{22}), \\ M_4 &= a_{22}(b_{21} - b_{11}), \\ M_5 &= (a_{11} + a_{12})b_{22}, \\ M_6 &= (a_{21} - a_{11})(b_{11} + b_{12}), \\ M_7 &= (a_{12} - a_{22})(b_{21} + b_{22}), \end{aligned}$$

we obtain the formula

$$\begin{aligned}
M_{\langle 2 \rangle} = & (a_{11}^* + a_{22}^*) \otimes (b_{11}^* + b_{22}^*) \otimes (c_{11} + c_{22}) \\
& + (a_{21}^* + a_{22}^*) \otimes b_{11}^* \otimes (c_{21} - c_{22}) \\
& + a_{11}^* \otimes (b_{12}^* - b_{22}^*) \otimes (c_{12} + c_{22}) \\
& + a_{22}^* \otimes (b_{21}^* - b_{11}^*) \otimes (c_{21} + c_{11}) \\
& + (a_{11}^* + a_{12}^*) \otimes b_{22}^* \otimes (c_{12} - c_{11}) \\
& + (a_{21}^* - a_{11}^*) \otimes (b_{11}^* + b_{12}^*) \otimes c_{22} \\
& + (a_{12}^* - a_{22}^*) \otimes (b_{21}^* + b_{22}^*) \otimes c_{11}.
\end{aligned}$$

Hence, $\mathbf{R}(M_{\langle 2 \rangle}) \leq 7$.

- (3) In fact, there is no algorithm for 2×2 matrix multiplication with less than 6 multiplications, furthermore $\underline{\mathbf{R}}(M_{\langle 2 \rangle}) = 7$ by [Landsberg, 2006].
- (4) For the 3×3 matrix multiplication tensor $M_{\langle 3 \rangle}$,

$$\mathbf{R}(M_{\langle 3 \rangle}) \leq 3^3 = 27,$$

$$\mathbf{R}(M_{\langle 3 \rangle}) \leq 23 \text{ (by Laderman 1976),}$$

$$19 \leq \mathbf{R}(M_{\langle 3 \rangle}) \leq 23 \text{ (by Stother's thesis 2010),}$$

$$\underline{\mathbf{R}}(M_{\langle 3 \rangle}) \leq 21 \text{ (by Schönhage),}$$

$$14 \leq \underline{\mathbf{R}}(M_{\langle 3 \rangle}) \text{ (by Bläser),}$$

$$15 \leq \underline{\mathbf{R}}(M_{\langle 3 \rangle}) \text{ (by Landsberg and Ottaviani 2012).}$$

As n grows, the total number of operations (i.e., additions and multiplications) for the $n \times n$ matrix multiplication tensor $M_{\langle n \rangle}$ approximates to the number of multiplications.

Definition 3.2 (The exponent of matrix multiplication). The number

$$\omega := \inf_{\tau} \{ \tau : M_{\langle n \rangle} \text{ can be computed with } O(n^{\tau}) \text{ operations} \}$$

is called the exponent of matrix multiplication.

The number ω is a crucial measure of complexity. The followings are world records for ω :

$$\begin{aligned}
\omega & \leq 3 \text{ (by the standard algorithm)} \\
& \leq \log_2 7 = 2.81... \text{ (by Strassen's algorithm)} \\
& \leq 2.7799... \text{ (1979)} \\
& \leq 2.48... \text{ (1987)} \\
& \leq 2.375477... \text{ (1990)} \\
& \leq 2.3736... \text{ (2010)} \\
& \leq 2.37294... \text{ (2012)} \\
& \leq 2.3728596... \text{ (2020)} \\
& \leq 2.371339... \text{ (2024)}
\end{aligned}$$

Conjecture 3.3. $\omega = 2$.

4. Symmetric Tensor and Symmetric Tensor Rank

Let $V_1, \dots, V_d$ be vector spaces. Consider the case $V_1 = V_2 = \dots = V_d =: V$. (Note that our base field $\mathbb{K}$ is of $\text{char}(\mathbb{K}) = 0$.) Let $V^{\otimes d}$ denote $\underbrace{V \otimes \dots \otimes V}_{d\text{-times}}$.

4.1. Symmetric tensors

Then there exists a natural *symmetrization* operator

$$\pi_S : V^{\otimes d} \rightarrow V^{\otimes d}, v_1 \otimes \dots \otimes v_d \mapsto \frac{1}{d!} \sum_{\sigma \in \mathfrak{S}_d} \underbrace{\sigma(v_1 \otimes \dots \otimes v_d)}_{=v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(d)}}.$$

The image $S^d V := \pi_S(V^{\otimes d})$ is a subspace of $V^{\otimes d}$ which consists of “symmetric tensors”.

Definition 4.1 (Symmetric tensor). Let $T \in V^{\otimes d}$. The tensor T is said to be *symmetric* if T is invariant under π_S , i.e., $\pi_S(T) = T$.

Example 10. (1) $e_0 \otimes e_1 + e_1 \otimes e_0$ is symmetric.

(2) $e_0 \otimes e_1 + e_1 \otimes e_1$ is not symmetric.

(3) The W-state tensor $W = e_0 \otimes e_0 \otimes e_1 + e_0 \otimes e_1 \otimes e_0 + e_1 \otimes e_0 \otimes e_0$ is symmetric.

Exercise 8. Show that $S^d V (\subset V^{\otimes d})$ is invariant under the $GL(V)$ -action.

Exercise 9. Let $\dim(V) = n + 1$. Show that $\dim(S^d V) = \binom{n+d}{d}$.

Remark 4.2 (Polarization). Recall that $(V^*)^{\otimes d} = \text{Hom}(V^{\otimes d}, \mathbb{K}) = \{d\text{-linear functions } V \times \dots \times V \rightarrow \mathbb{K}\}$. Consider $S^d V^*$ a subspace of $(V^*)^{\otimes d}$. Obviously, $(V^*)^{\otimes d}$ is identified with the set of symmetric d -linear functions $V \times \dots \times V \rightarrow \mathbb{K}$. We will check some relations between these sets and the set of homogeneous polynomial of degree d with variables in V^* and the zero polynomial. Let $S_d V^*$ denote the set of homogeneous polynomial of degree d with variables in V^* and the zero polynomial.

(1) There is a one-to-one correspondence between $S^d V^*$ and $S_d V^*$. For example, the W-state tensor $W = e_0 \otimes e_0 \otimes e_1 + e_0 \otimes e_1 \otimes e_0 + e_1 \otimes e_0 \otimes e_0 \in \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ corresponds to $3xy^2$.

(2) There is a one-to-one correspondence between $S_d V^*$ and the set of symmetric d -linear functions $V \times \dots \times V \rightarrow \mathbb{K}$, by the *polarization* and *diagonalization*. When $d = 2$, the polarization of $g \in S_d V^*$ is given as

$$g(\mathbf{x}) \mapsto Q(\mathbf{x}_1, \mathbf{x}_2) := \frac{1}{2!} \{g(\mathbf{x}_1 + \mathbf{x}_2) - g(\mathbf{x}_1) - g(\mathbf{x}_2)\},$$

and the diagonalization of Q is given as $Q(\mathbf{x}, \mathbf{x}) \in S_2 V^*$. For example, the polarization of $g(\mathbf{x}) = g(x, y) = x^2 + 3xy + 2y^2$ is

$$\begin{aligned}
 Q(\mathbf{x}_1, \mathbf{x}_2) &:= Q((x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)})) \\
 &= \frac{1}{2} \{g(x^{(1)} + x^{(2)}, y^{(1)} + y^{(2)}) - g(x^{(1)}, y^{(1)}) - g(x^{(2)}, y^{(2)})\} \\
 &= \frac{1}{2} \{(x^{(1)} + x^{(2)})^2 + 3(x^{(1)} + x^{(2)})(y^{(1)} + y^{(2)}) + 2(y^{(1)} + y^{(2)})^2\} \\
 &\quad - \frac{1}{2} \{(x^{(1)})^2 + 3x^{(1)}y^{(1)} + 2(y^{(1)})^2\} \\
 &\quad - \frac{1}{2} \{(x^{(2)})^2 + 3x^{(2)}y^{(2)} + 2(y^{(2)})^2\} \\
 &= x^{(1)}x^{(2)} + \frac{3}{2}x^{(2)}y^{(1)} + \frac{3}{2}x^{(1)}y^{(2)} + 2y^{(1)}y^{(2)}.
 \end{aligned}$$

For general $d \in \mathbb{Z}_{\geq 1}$, the polarization of a homogeneous polynomial f of degree d is given as

$$F(\mathbf{x}_1, \dots, \mathbf{x}_d) = \frac{1}{d!} \sum_{\substack{I \neq \emptyset \\ I \subset [d]}} (-1)^{d-|I|} F \left(\sum_{i \in I} \mathbf{x}_i \right)$$

where $\mathbf{x}_i = (x_1^{(i)}, \dots, x_n^{(i)})$ for each $i = 1, \dots, d$ and $|I|$ denotes the number of elements in the set I , and the diagonalization of F is given as $F(\mathbf{x}, \dots, \mathbf{x}) \in S_d V^*$.

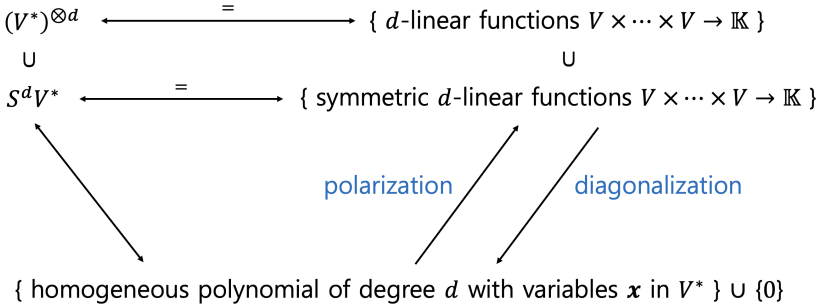


FIGURE 5. Polarization and diagonalization.

4.2. Alternating tensors

Definition 4.3 (Alternating tensor). Define a map

$$\pi_\Lambda : V^{\otimes d} \rightarrow V^{\otimes d}, v_1 \otimes \dots \otimes v_d \mapsto \frac{1}{d!} \sum_{\sigma \in \mathfrak{S}_d} (\text{sgn } \sigma) v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(d)} =: v_1 \wedge \dots \wedge v_d.$$

The image $\pi_\Lambda(V^{\otimes d}) \subset V^{\otimes d}$ is called the subspace of alternating (or skew-symmetric) d -tensors.

Note that

$$\pi_\Lambda(V^{\otimes d}) = \Lambda^d V := \{T \in V^{\otimes d} : \sigma \cdot T = (\text{sgn } \sigma)T \text{ for all } \sigma \in \mathfrak{S}_d\}.$$

Exercise 10. (1) Let $v_1, \dots, v_k \in V$. Show that $v_1 \wedge \dots \wedge v_k = 0$ if and only if $v_1, \dots, v_k$ are linearly dependent.

(2) Show that $V^{\otimes 2} = S^2 V \oplus \Lambda^2 V$.

(3) Show that $\Lambda^k V$ is invariant under the $GL(V)$ -action.

Remark 4.4. $V^{\otimes 3} \neq S^3 V \oplus \Lambda^3 V$. Partially symmetric (or partially skew-symmetric) tensors are in $V^{\otimes 3}$.

4.3. Symmetric tensor rank

Definition 4.5 (Symmetric tensor rank and symmetric border rank). For a given $\phi \in S^d V$, we define

$$\mathbf{R}_S(\phi) := \min\{r : \phi = v_1^d + \dots + v_r^d \text{ for some } v_i \in V\}$$

and call it the *symmetric tensor rank* of ϕ . Here, the expression $\phi = v_1^d + \dots + v_r^d$ is called a *symmetric tensor decomposition*. Similarly, we define

$$\underline{\mathbf{R}}_S(\phi) := \min \left\{ r : \begin{array}{l} \text{there exists a family of symmetric tensors } \{\phi_\epsilon\} \\ \text{such that } \mathbf{R}_S(\phi_\epsilon) = r \text{ for all } \epsilon \text{ and } \lim_{\epsilon \rightarrow 0} \phi_\epsilon = \phi \end{array} \right\}$$

and call it the *symmetric (tensor) border rank* of ϕ .

Remark 4.6. In terms of (homogeneous) polynomial, the above can be stated: for $F \in S_d V$,

$$\mathbf{R}_S(F) = \min\{r : F = L_1^d + \dots + L_r^d, L_i \in S_1 V\}.$$

We often call it the *Waring rank* of F , and call the expression $F = L_1^d + \dots + L_r^d$ a *power sum decomposition* or *Waring decomposition* of F .

Remark 4.7. For any symmetric tensor $T \in S^d V \subset V^{\otimes d}$, we obviously obtain $\mathbf{R}(T) \leq \mathbf{R}_S(T)$. We can naturally ask “Is $\mathbf{R}(T) < \mathbf{R}_S(T)$ possible.” Comon’s conjecture says that $\mathbf{R}(T) = \mathbf{R}_S(T)$ may be true. It is still open.

Example 11. (1) Consider the monomial $xy \in \mathbb{K}[x, y]$. Since $xy = \frac{1}{4}(x + y)^2 - \frac{1}{4}(x - y)^2$, then $\mathbf{R}_S(xy) \leq 2$. Moreover, xy cannot be l^2 for some linear form l , because $\mathbb{K}[x, y]$ is UFD.

(2) Consider the monomial $xyz \in \mathbb{K}[x, y, z]$. Since $xy = \frac{1}{24}\{(x + y + z)^3 - (x - y + z)^3 - (z + y - z)^3 + (x - y - z)^3\}$, then $\mathbf{R}_S(xyz) \leq 4$. Moreover, since $\mathbb{K}[x, y, z]$ is UFD, then $\mathbf{R}_S(xyz) \geq 2$. (In fact, $\mathbf{R}_S(xyz) = 4$.)

(3) Consider $x^2 y \in \mathbb{K}[x, y]$. Since $6x^2 y = (x + y)^3 - (x - y)^3 - 2y^3 = (x + y)^3 + (y - x)^3 + (-\sqrt[3]{2}y)^3$, then $\mathbf{R}_S(x^2 y) \leq 3$. In fact, $\mathbf{R}(x^2 y) = 3$ because $x^2 y$ corresponds to the W-state tensor. Thus, $\mathbf{R}_S(x^2 y) = 3$.

(4) (B. Reznick) $\mathbf{R}_S((x^2 + y^2 + z^2)^2) = 6$ over $\mathbb{R}$. Furthermore, $(x^2 + y^2 + z^2)^2 = \sum_{i=1}^6 (a_i x + b_i y + c_i z)^4$ if and only if $\{\pm(a_i, b_i, c_i) : i = 1, \dots, 6\}$ are the vertices of an icosahedron.

- (5) For the 2×2 determinant polynomial $\det_2 = xy - zw$, $\mathbf{R}_S(\det_2) = 4$. In addition, for the 3×3 determinant polynomial $\det_3$, $15 \leq \mathbf{R}_S(\det_3) \leq 18$, where the lower bound is from [Boij-Teitler, 2020] and the upper bound is from [Conner-Gesmundo-Landsberg-Ventura, 2019].

Remark 4.8. (1) The problem to find the tensor rank (or symmetric tensor rank) of a given tensor (resp. symmetric tensor) is an NP-hard problem.
 (2) The existence of a symmetric tensor decomposition for a given symmetric tensor holds, since the set of power sum of linear forms is not included in a hyperplane of $S_d V$. (See [Comon-Golub-Lim-Mourrain, 2008].)

Example 12 (Forms with known Waring rank). Consider forms in $S_d V$ where $\dim V = n$.

- (1) Quadratic forms(i.e., $d = 2$): For a quadratic form Q , its symmetric tensor rank $\mathbf{R}_S(Q)$ is the same with the rank of the corresponding symmetric matrix.
- (2) Binary forms(i.e., $n = 2$): By Sylvester algorithm (which we will study in Section 5), we can calculate its symmetric tensor rank.
- (3) $d = 3, n = 3, 4$: by [B. Segre].
- (4) $d = 4, n = 3$: by [J. Kleppe, 1999]
- (5) General forms: by [Alexander-Hirschowitz, 1995]
- (6) Monomials: Over $\mathbb{C}$, $\mathbf{R}_S(x_0^{a_0} x_1^{a_1} \cdots x_n^{a_n}) = (a_0 + 1) \cdots (a_{n-1} + 1)$ where $a_0 \geq a_1 \geq \cdots \geq a_n$, by [Carlini-Catalisano-Geramita, 2012] and [Buczyńska-Buczyński-Teitler, 2012]. Over $\mathbb{R}$, some results are known, by [Boij-Carlini-Geramita, 2012], [Carlini-Kummer-Oneto-Ventura, 2017], and [Han-Moon, 2020].
- (7) Some scattered examples.

5. Lower Bound of Symmetric Tensor Rank

In our lecture, $\mathbb{K}$ is a field of characteristic 0. Let $S = \mathbb{K}[x_0, \dots, x_n]$ and $T = \mathbb{K}[\partial_0, \dots, \partial_n]$ where $\partial_i := \frac{\partial}{\partial x_i}$ for $i = 0, \dots, n$.

5.1. Differentiation action and perp ideal

We naturally obtain the differentiation action $\lrcorner : T \curvearrowright S$, for instance, $\partial_i \lrcorner x_i^3 = 3x_i^2$. For any $F \in S_d$, we define

$$F^\perp := \{\theta \in T : \theta \lrcorner F = 0\}$$

and call it the *perp* (or *apolar* or *annihilating*) *ideal* of F . Then $F^\perp$ is an ideal. Moreover, it is *Artinian Gorenstein* ideal in T .

Exercise 11. Let $F \in \mathbb{K}[x, y]_d$ and let $\alpha = \frac{\partial}{\partial x}, \beta = \frac{\partial}{\partial y}$.

- (1) Suppose that $t_1, \dots, t_{d+1} \in \mathbb{K}$ are any pairwise distinct numbers. Show that $(x + t_1 y)^d, \dots, (x + t_{d+1} y)^d$ are linearly independent and span the vector space $\mathbb{K}[x, y]_d$.

- (2) Let $F = \lambda_1(a_1x + b_1y)^d + \cdots + \lambda_r(a_rx + b_ry)^d \in \mathbb{K}[x, y]_d$ where $a_i, b_i \in \mathbb{K}$ for $i = 1, \dots, r$. Let $h(\alpha, \beta) = (b_1\alpha - a_1\beta) \cdots (b_r\alpha - a_r\beta) \in T$. Show that $h \in F^\perp$.
- (3) Conversely, suppose that $h(\alpha, \beta) (\neq 0) \in F^\perp$ with r distinct roots. Show that there exist $\lambda_1, \dots, \lambda_r \in \mathbb{K}$ such that $F = \sum_{i=1}^r \lambda_i(a_ix + b_iy)^d$.

From above exercise, we obtain the following theorem.

Theorem 5.1 (Sylvester's algorithm). *Let $F \in \mathbb{K}[x, y]_d$. Then*

$\mathbf{R}_S(F) = \min\{r : F^\perp \text{ contains a form } f \text{ of degree } r \text{ with (projectively) distinct roots}\}.$

Example 13. Let $F = x^3y \in \mathbb{K}[x, y]_4$. Then

$$F^\perp = (\alpha^4, \beta^2)$$

in $T = \mathbb{K}[\alpha, \beta]$, where $\alpha = \frac{\partial}{\partial x}$ and $\beta = \frac{\partial}{\partial y}$.

- (i) (degree 1) $(F^\perp)_1 = 0$.
- (ii) (degree 2) $(F^\perp)_2 = (\beta^2)$ contains only forms $\lambda\beta^2$ ($\lambda \in \mathbb{K}$) with multiple roots.
- (iii) (degree 3) $(F^\perp)_3 = (\alpha\beta^2, \beta^3)$ contains only forms $\lambda\alpha\beta^2 + \mu\beta^3 = \beta^2(\lambda\alpha + \mu\beta)$ with multiple roots.
- (iv) (degree 4) $(F^\perp)_4 = (\alpha^4, \alpha^2\beta^2, \alpha\beta^3, \beta^4)$ contains the form

$$(4\alpha^2 - \beta^2)(\alpha^2 - 4\beta^2) = (2\alpha + \beta)(2\alpha - \beta)(\alpha + 2\beta)(\alpha - 2\beta)$$

with distinct roots.

Thus, $\mathbf{R}_S(x^3y) = 4$.

5.2. Apolarity lemma

Exercise 12. Let $F \in S_d$. Let $P = (\alpha_0, \dots, \alpha_n)$ be a point in $\mathbb{K}^{n+1}$ and $D_P = \alpha_0\partial_0 + \cdots + \alpha_n\partial_n \in T_1, l_P = \alpha_0x_0 + \cdots + \alpha_nx_n \in S_1$.

- (1) Show that $D_P^d \lrcorner F = d! \cdot F(P)$.
- (2) Let $D \in T_k$. Show that, for all $d \geq k$, $D([P]) = 0$ (considering D as a polynomial) if and only if $D \lrcorner l_P^d = 0$.
- (3) Show that $D_P^k \in F^\perp$ if and only if F vanishes to order (at least) $d+1-k$ at P (i.e., every partial derivative of F of order $\leq d-k$ vanishes at P).

Theorem 5.2 (Apolarity lemma). *Let $\mathbb{K}$ be algebraically closed field with characteristic 0. Let $F \in S_d, l_1, \dots, l_r \in S_1$, and let $\mathbb{X} = \{[l_1], \dots, [l_r]\} \subset \mathbb{P}(S_1)$. Assume that $[l_i]$'s are distinct. Then there exist $\lambda_1, \dots, \lambda_r \in \mathbb{K}$ such that $F = \lambda_1 l_1^d + \cdots + \lambda_r l_r^d$ if and only if $I_{\mathbb{X}} := \{D \in T : D([l_i]) = 0\} \subset F^\perp$.*

Remark 5.3. The apolarity lemma may give us some upper bounds of $\mathbf{R}_S(F)$ for a given $F \in S_d$.

Example 14 (Waring rank of monomials over $\mathbb{C}$). Let $F = x_0^{a_0} x_1^{a_1} \cdots x_n^{a_n} \in \mathbb{C}[x_0, \dots, x_n]$ where $0 < a_0 \leq \cdots \leq a_n$ and $d = \sum_{i=0}^n a_i$. Then it is easy to check that

$$F^\perp = (\partial_0^{a_0+1}, \partial_1^{a_1+1}, \dots, \partial_n^{a_n+1}).$$

Take

$$I = (\partial_1^{a_1+1} - \phi_1 \cdot \partial_0^{a_0+1}, \partial_2^{a_2+1} - \phi_2 \cdot \partial_0^{a_0+1}, \dots, \partial_n^{a_n+1} - \phi_n \cdot \partial_0^{a_0+1})$$

for some $\phi_i \in \mathbb{C}[\partial_0, \dots, \partial_n]_{a_i - a_0}$, $i = 1, \dots, n$. Then I is a *complete intersection ideal* of points of degree $(a_1 + 1)(a_2 + 1) \cdots (a_n + 1)$ such that

$$I \subset F^\perp.$$

By the apolarity lemma, we obtain $\mathbf{R}_S(x_0^{a_0} \cdots x_n^{a_n}) \leq (a_1 + 1) \cdots (a_n + 1)$.

The exact symmetric tensor rank of $(x_0^{a_0} \cdots x_n^{a_n})$ was discovered by [Carlini-Catalisano-Geramita, 2012]. Furthermore, by [Buczyńska-Buczyński-Teitler, 2012], it is proved that its minimal apolar set always comes from a complete intersection ideal.

Exercise 13. For $F = x_0^{a_0} x_1^{a_1} \cdots x_n^{a_n} \in \mathbb{K}[x_0, \dots, x_n]$, show that

$$F^\perp = (\partial_0^{a_0+1}, \partial_1^{a_1+1}, \dots, \partial_n^{a_n+1}).$$

5.3. Sylvester's Bound

Definition 5.4. For $F \in S_d$, we let $\text{Derivs}(F)$ denote the vector space spanned by all derivatives of F of all orders (including F itself), and let $\text{Derivs}(F)_a$ denote the vector space spanned by all derivatives of F of order $d - a$.

Example 15. For $xyz \in \mathbb{K}[x, y, z]$, $\text{Derivs}(xyz) = \langle xyz, xy, yz, xz, x, y, z, 1 \rangle$.

Proposition 5.5 (Sylvester's bound). For all $a \in \mathbb{Z}_{\geq 0}$, $\mathbf{R}_S(F) \geq \dim(\text{Derivs}(F)_a)$. In particular, $\mathbf{R}_S(F) \geq \max_{0 \leq a \leq d} \{\dim(\text{Derivs}(F)_a)\}$.

Proof. Say $F = l_1^d + \cdots + l_r^d$ with $r = \mathbf{R}_S(F)$. Then

$$\frac{\partial F}{\partial x} = d \cdot l_1^{d-1} \cdot c_1 + \cdots + d \cdot l_r^{d-1} \cdot c_r \in \langle l_1^{d-1}, \dots, l_r^{d-1} \rangle,$$

where c_i is the coefficient of x on l_i for $i = 1, \dots, r$. Similarly, we obtain $\text{Derivs}(F)_a \subset \langle l_1^a, \dots, l_r^a \rangle$. Hence, $\dim(\text{Derivs}(F)_a) \leq \dim(\langle l_1^a, \dots, l_r^a \rangle) \leq r$. $\square$

Example 16. Consider the monomial $xyz \in \mathbb{K}[x, y, z]$. Since

$$\text{Derivs}(xyz) = \langle xyz, xy, yz, xz, x, y, z, 1 \rangle,$$

then $\dim(\text{Derivs}(xyz)_0) = 1$, $\dim(\text{Derivs}(xyz)_1) = 3$, $\dim(\text{Derivs}(xyz)_2) = 3$, $\dim(\text{Derivs}(xyz)_3) = 1$. Thus, $\mathbf{R}_S(xyz) \geq 3$.

Example 17. Consider the $d \times d$ determinant polynomial $\det_d$. Since $\text{Derivs}(\det_d)$ is the vector space spanned by all minors (i.e., determinants of submatrices), then

$$\mathbf{R}_S(\det_d) \geq \binom{d}{\lfloor d/2 \rfloor}^2.$$

For instance, $\mathbf{R}_S(\det_3) \geq 3^2 = 9$.

6. Polynomial Waring Problem

6.1. Waring problem

Problem 6.1 (Waring problem). Find the minimum $r \in \mathbb{Z}_{\geq 0}$ such that a general form (or a specific form) $F \in S_d$ can be written as the sum of at most r d -th power of linear forms.

Let $S = \mathbb{K}[x_0, \dots, x_n]$. Consider the map

$$\phi_{d,r} : \underbrace{S_1 \times \dots \times S_1}_{r\text{-times}} \rightarrow S_d, (L_1, \dots, L_r) \mapsto L_1^d + \dots + L_r^d.$$

- (1) To answer Waring problem for all forms is equivalent with the problem: find the minimum $r \in \mathbb{Z}_{\geq 0}$ such that $\text{Im}(\phi_{d,r}) = S_d$.
- (2) To answer Waring problem for general forms is equivalent with the problem: find the minimum $r \in \mathbb{Z}_{\geq 0}$ such that $\overline{\text{Im}(\phi_{d,r})} = S_d$, i.e., $\dim(\text{Im}(\phi_{d,r})) = \dim(S_d)$.

Note that $\phi_{d,r}$ is a *polynomial map* from an affine space $\mathbb{A}^{r(n+1)}$ to an affine space $\mathbb{A}^N$ for $N := \binom{n+d}{d}$.

6.2. Alexander-Hirschowitz theorem

In this section, we will interpret the Waring problem geometrically. Consider the map

$$\nu_d : \mathbb{P}^n = \mathbb{P}(S_1) \rightarrow \mathbb{P}(S_d), [L] \mapsto [L^d]$$

which is called the *(d-th) Veronese map*. Then

$$\widehat{\nu_d(\mathbb{P}^n)} = \text{Im}(\phi_{d,1}) = \overline{\text{Im}(\phi_{d,1})},$$

where the last equality holds since $\text{Im}(\phi_{d,1})$ is Zariski closed (we call $\nu_d(\mathbb{P}^n)$ the *d-th Veronese variety*). Since what we want to investigate is $\overline{\text{Im}(\phi_{d,r})}$ in the Waring problem for general forms, we want to express $\overline{\text{Im}(\phi_{d,r})}$ in terms of the image of Veronese map. For this, we will use the notion of *secant variety*.

Definition 6.2 (Secant variety). Let $X \subset \mathbb{P}^N$ be a *nondegenerate* (i.e., which is not contained in a hyperplane) irreducible variety. Then

$$\sigma_r(X) := \overline{\bigcup_{p_1, \dots, p_r \in X} \langle p_1, \dots, p_r \rangle} \subset \mathbb{P}^N$$

is called the *r-th secant variety* of X . In addition, we let

$$\sigma_r^0(X) := \bigcup_{p_1, \dots, p_r \in X} \langle p_1, \dots, p_r \rangle.$$

Remark 6.3. Let $X \subset \mathbb{P}^N$ be a *nondegenerate* (i.e., which is not contained in a hyperplane) irreducible variety. We easily obtain the following properties of secant variety.

- (1) $\sigma_1(X) = X$.
- (2) $X = \sigma_1(X) \subsetneq \sigma_2(X) \subsetneq \sigma_3(X) \subsetneq \dots \subsetneq \sigma_n(X) = \mathbb{P}^N$ for some n .

(3) When $\dim(X) = n$, the obtain the *expected diemnsion* of $\sigma_r(X)$ as

$$\min\{N, r \cdot n + (r - 1)\}$$

by counting parameters. If $\dim(\sigma_r(X)) < \min\{N, r \cdot n + (r - 1)\}$, then $\sigma_r(X)$ is said to be *r-defective*.

(4) $\text{Im}(\phi_{d,r}) = \overline{\sigma_r^0(\nu_d(\mathbb{P}S_1))}$. In particular, $\overline{\text{Im}(\phi_{d,r})} = \overline{\sigma_r(\nu_d(\mathbb{P}S_1))}$, and so

$$\dim(\text{Im}(\phi_{d,r})) = \dim(\overline{\sigma_r(\nu_d(\mathbb{P}S_1))}).$$

Remark 6.4 (Expectation over $\mathbb{C}$). Let $\mathbb{K} = \mathbb{C}$, and let $F \in S_d$ so that

$$F = \sum_{\substack{i_0 + \dots + i_n = d \\ i_k \geq 0}} a_{i_0, \dots, i_n} x_0^{i_0} \cdots x_n^{i_n} \in \mathbb{C}[x_0, \dots, x_n]_d.$$

We want a Waring rank decomposition

$$F = \sum_{i=1}^r (b_{i,0}x_0 + \cdots + b_{i,n}x_n)^d.$$

Then we have the system of polynomials:

$$a_{i_0, \dots, i_n} = f_{i_0, \dots, i_n}(b_{1,0}, \dots, b_{r,n})$$

for some polynomials $f_{i_0, \dots, i_n}$ of $r(n+1)$ variables, for $i_0, \dots, i_n \in \mathbb{Z}_{\geq 0}$ such that $i_0 + \cdots + i_n = d$. This system consists of $\binom{n+d}{d}$ equations with $r(n+1)$ variables. Hence, for solvability of general system of this type, we expect the condition that

$$r(n+1) \geq \binom{n+d}{d}, \text{ i.e., } r \geq \left\lceil \frac{1}{n+1} \binom{n+d}{d} \right\rceil.$$

Now, we have a natural question: Are there exceptional cases? The Alexander-Hirschowitz theorem classified all exceptional cases.

Theorem 6.5 (Alexander-Hirschowitz theorem). *Let $\mathbb{K} = \mathbb{C}$. Let F be a general form in $\mathbb{C}[x_0, \dots, x_n]_d$. Then $\mathbf{R}_S(F) = \left\lceil \frac{1}{n+1} \binom{n+d}{d} \right\rceil$ except for the following cases:*

- $d = 2, n \geq 2$ (in this case, $\mathbf{R}_S(F) = n + 1$, not $\frac{n+2}{2}$);
- $d = 3, n = 4$ (in this case, $\mathbf{R}_S(F) = 8$, not 7);
- $d = 4, n = 2$ (in this case, $\mathbf{R}_S(F) = 6$, not 5);
- $d = 4, n = 3$ (in this case, $\mathbf{R}_S(F) = 10$, not 9);
- $d = 4, n = 4$ (in this case, $\mathbf{R}_S(F) = 15$, not 14).

7. Secant Variety

In this section, let $\mathbb{K} = \mathbb{C}$.

7.1. Join and secant

In this section, we review about join and secant variety. Let $X, Y \subseteq \mathbb{P}^N$ be projective varieties.

- (1) The *join* of X and Y is

$$J(X, Y) := \overline{\bigcup_{\substack{p \in X, \\ q \in Y}} \langle p, q \rangle} \subseteq \mathbb{P}^N.$$

- (i) The expected dimension of $J(X, Y)$ is $\dim(X) + \dim(Y) + 1$.
 (ii) $\widehat{J(X, Y)} = \widehat{X} + \widehat{Y}$, where the sum on the right-hand side is the sum of affine varieties, i.e., the Zariski closure of the image of the addition map $+: \mathbb{A}^{N+1} \times \mathbb{A}^{N+1} \rightarrow \mathbb{A}^{N+1}, (v, w) \mapsto v + w$.
 (iii) The set of projective varieties of $\mathbb{P}^N$ becomes an ordered abelian monoid with respect to the join operation where the unit is the empty variety.
- (2) The *k-th secant variety* of $X \subseteq \mathbb{P}^N$ is

$$\sigma_k(X) = \underbrace{J(X \cdots J(X, J(X, X)))}_{k\text{-times}} = \overline{\bigcup_{x_1, \dots, x_k \in X} \langle x_1, \dots, x_k \rangle} \subseteq \mathbb{P}^N.$$

- (i) The expected dimension of $\sigma_k(X)$ is $k \cdot \dim(X) + (k - 1)$. Note that the actual dimension may be strictly smaller than the expected dimension, as we checked from the Alexander-Hirschowitz theorem.
 (ii) We call the value

$$\min\{N, k \cdot \dim(X) + (k - 1)\} - \dim(\sigma_k(X))$$

the *secant defect* of $\sigma_k(X)$. If secant defect is 0, then $\sigma_k(X)$ is said to be *non-defective*, and if secant defect is positive, then $\sigma_k(X)$ is said to be *defective*.

- (3) We can define the secant variety in terms of incidence variety. The set

$$I := \overline{\{(a, x_1, \dots, x_k) : a \in \langle x_1, \dots, x_k \rangle, \dim(\langle x_1, \dots, x_k \rangle) = k - 1\}}$$

in $\mathbb{P}^N \times \underbrace{X \times \cdots \times X}_{k\text{-times}}$ is called an *incidence variety*.

- (i) $\dim(I) = k \cdot \dim(X) + (k - 1)$.
 (ii) Consider the projection maps $p : I \rightarrow \mathbb{P}^N$, $q : I \rightarrow X \times \cdots \times X$, and $\pi_i : X \times \cdots \times X \rightarrow X$ (into the i -th factor) for $i = 1, \dots, k$. (See Figure 6). We have

$$\sigma_k(X) = p(I).$$

In fact, $\dim(\sigma_k(X))$ is related to fibers of p . In more detail, if

$$\dim(p^{-1}(a)) > 0$$

for a general $a \in \sigma_k(X)$, then $\sigma_k(X)$ is defective. In addition, a subvariety $\pi_i(q(p^{-1}(a)))$ of X for a general $a \in \sigma_k(X)$ is also related to $\dim(\sigma_k(X))$.

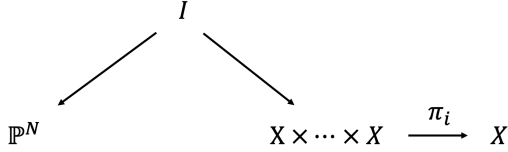


FIGURE 6.

- (iii) From the incidence space I^0 defined by not taking the Zariski closure in the definition of I , we can also check the *identifiability* of a tensor. Let X be the space of rank one tensors, i.e., *Segre variety*. Assume that, for $T \in \sigma_k^0(X) = p(I^0)$, the number of elements in $p^{-1}(T)$ is one. Then the corresponding tensor T is *identifiable*, i.e., the tensor decomposition with k summands is unique.

7.2. Segre and Veronese embeddings

In this subsection, we review about Segre and Veronese embeddings. Let $V_1, \dots, V_d$ and V be vector spaces.

- (1) The map

$$\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d) \rightarrow \mathbb{P}(V_1 \otimes \cdots \otimes V_d), ([v_1], \dots, [v_d]) \mapsto [v_1 \otimes \cdots \otimes v_d]$$

is called a *Segre embedding*. The image of this map, denoted by $\text{Seg}(\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d))$, is called a *Segre variety*. This map parametrizes all rank one tensors in $\mathbb{P}(V_1 \otimes \cdots \otimes V_d)$.

- (2) The map

$$\mathbb{P}(V) \rightarrow \mathbb{P}(S^d V), [v] \mapsto [v^d := \underbrace{v \otimes \cdots \otimes v}_{d\text{-times}}]$$

is called the *d-th Veronese embedding*. The image of this map, denoted by $\nu_d(\mathbb{P}(V))$, is called the *d-th Veronese variety*. This map parametrizes all symmetric rank one tensors in $\mathbb{P}(S^d V)$.

The following proposition interprets the algebraic notions border rank and symmetric border rank geometrically in terms of secant variety.

Proposition 7.1. (Recall that $\mathbb{K} = \mathbb{C}$ in this section.)

- (1) The k -th secant variety of $\text{Seg}(\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d))$ is the set of all tensors in $\mathbb{P}(V_1 \otimes \cdots \otimes V_d)$ of border rank $\leq k$.
- (2) The k -th secant variety of $\nu_d(\mathbb{P}(V))$ is the set of all symmetric tensors in $\mathbb{P}(S^d V)$ of symmetric border rank $\leq k$.

7.3. Dimension of secant variety of Segre varieties

Consider a Segre variety $\text{Seg}(\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d))$ where $\dim(\mathbb{P}(V_i)) = n_i$ for each $i = 1, \dots, d$. Assume that $n_1 \leq n_2 \leq \cdots \leq n_d$. Since $\dim(\text{Seg}(\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d))) = n_1 + \cdots + n_d$, then the expected dimension of $\sigma_k(\text{Seg}(\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d)))$ is $k \cdot (n_1 + \cdots + n_d) + (k - 1)$.

From now on, we denote $\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}$ instead of $\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_d)$.

Theorem 7.2 ([Catalisano-Germita-Gimigliano, 2002]). *Let $d \geq 3$. $\sigma_k(\text{Seg}(\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}))$ is non-defective if*

- (1) $k \leq n_1 + 1$, or
- (2) $\max\{n_d + 1, k\} \leq \frac{n_1 + \cdots + n_d + 1}{2}$.

Definition 7.3. Let $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{Z}_{\geq 1}^d$. $\mathbf{n}$ is said to be *unbalanced* if $n_d - 1 \geq \prod_{i=1}^{d-1} (n_i + 1) - \sum_{i=1}^{d-1} n_i$. We say $\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}$ is *unbalanced* if the corresponding d -tuple $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{Z}_{\geq 1}^d$ is unbalanced.

Theorem 7.4 ([Abo-Ottaviani-Peterson, 2009]). *If $\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}$ is unbalanced, then $\sigma_k(\text{Seg}(\mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}))$ is defective.*

Conjecture 7.5 ([Abo-Ottaviani-Peterson, 2009]). *Let $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_d}$. Then $\sigma_k(X)$ is defective only if*

- (i) X is unbalanced,
- (ii) $X = \mathbb{P}^2 \times \mathbb{P}^n \times \mathbb{P}^n$ for n even,
- (iii) $X = \mathbb{P}^2 \times \mathbb{P}^3 \times \mathbb{P}^3$, or
- (iv) $X = \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^n \times \mathbb{P}^n$ for n odd.

If this conjecture is true, then, in a balanced case $n_1 = \cdots = n_d$, there are only two defective cases $\sigma_4(\text{Seg}(\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^2))$ and $\sigma_3(\text{Seg}(\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1))$.

7.4. Maximal rank and generic Rank

Considering Proposition 7.1, we can generalize the notion of tensor rank (or symmetric tensor rank) by replacing Segre variety (resp. Veronese variety) by an arbitrary nondegenerate irreducible projective variety. Let X be a nondegenerate irreducible projective variety in $\mathbb{P}^N$. (See Figure 7.)

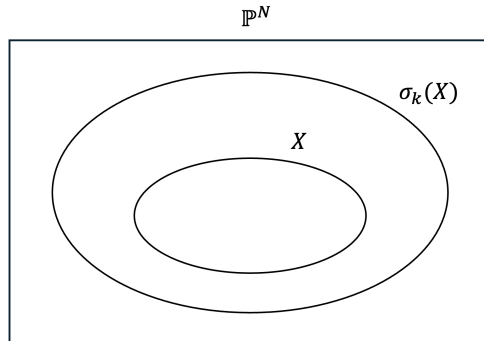


FIGURE 7.

Definition 7.6 (X -rank). Let $p \in \mathbb{P}^N$.

- (1) $\mathbf{R}_X(p) := \min\{r : p \in \sigma_r^0(X)\}$ is called the X -rank of p .
- (2) $\underline{\mathbf{R}}_X(p) := \min\{r : p \in \sigma_r(X)\}$ is called the X -border rank of p .

Definition 7.7 (Maximal rank). The smallest $r \in \mathbb{Z}_{\geq 1}$ such that $\sigma_r^0(X) = \mathbb{P}^N$ is called the *maximal rank* with respect to X .

Definition 7.8 (Generic rank). The smallest $r \in \mathbb{Z}_{\geq 1}$ such that $\sigma_r(X) = \mathbb{P}^N$ is called the *generic rank* with respect to X . In other words, $r \in \mathbb{Z}_{\geq 1}$ is the generic rank if the set $\{p \in \mathbb{P}^N : \mathbf{R}_X(p) = r\}$ contains a non-empty Zariski open (that is dense) subset of $\mathbb{P}^N$.

Theorem 7.9 ([Blekherman-Teitler, 2015]). Let $\mathbb{K}$ be algebraically closed, and let X be a nondegenerate irreducible projective variety in $\mathbb{P}^n$. Then

$$r_{\max} \leq 2 \cdot r_{\text{gen}}$$

where $r_{\max}$ and r_{gen} denote the maximal rank and generic rank with respect to X , respectively.

Theorem 7.10 ([Buczynski-Han-Mella-Teitler, 2016]). Let X be either Veronese variety or Segre variety. Then

$$r_{\max} \leq 2 \cdot r_{\text{gen}} - 1$$

where $r_{\max}$ and r_{gen} denote the maximal rank and generic rank with respect to X , respectively.